Supplementary Information

This document contains supplementary figures, tables, and other details supporting the main publication.

Supplementary Tables

Dataset	Input data	Accuracy	Precision	Recall	F1	AUC	PR AUC
Wolinska et al. n=175, 80/95	all OGs	0.47 ± 0.11	0.20 ± 0.34	0.33 ± 0.58	0.25 ± 0.43	0.50 ± 0.00	0.59 ± 0.00
	filtered OGs	0.41 ± 0.00	–	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00	0.59 ± 0.00
Finkel et al. 2020 n=44, 23/21	all OGs	0.51 ± 0.02	0.16 ± 0.28	0.33 ± 0.58	0.22 ± 0.38	0.50 ± 0.00	0.49 ± 0.00
	filtered OGs	0.50 ± 0.00	–	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00	0.50 ± 0.00
Hou et al. n=175, 71/104	all OGs	0.50 ± 0.01	0.17 ± 0.29	0.33 ± 0.58	0.22 ± 0.39	0.50 ± 0.00	0.51 ± 0.00
	filtered OGs	0.37 ± 0.00	–	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00	0.63 ± 0.00
Duran et al. n=86, 54/32	all KOs	0.53 ± 0.01	0.39 ± 0.01	0.46 ± 0.05	0.42 ± 0.03	0.55 ± 0.02	0.49 ± 0.03
	filtered KOs	0.57 ± 0.04	0.40 ± 0.04	0.30 ± 0.23	0.32 ± 0.14	0.57 ± 0.01	0.45 ± 0.03
	all OGs	0.57 ± 0.22	0.10 ± 0.18	0.33 ± 0.58	0.16 ± 0.27	0.50 ± 0.00	0.31 ± 0.00
	filtered OGs	0.63 ± 0.00	–	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00	0.37 ± 0.00
	ESM-2	0.48 ± 0.01	0.37 ± 0.02	0.54 ± 0.08	0.44 ± 0.04	0.54 ± 0.01	0.43 ± 0.05
	Bacformer	0.48 ± 0.01	0.36 ± 0.01	0.50 ± 0.00	0.42 ± 0.00	0.49 ± 0.03	0.38 ± 0.04
	DNABERT-S	0.50 ± 0.04	0.40 ± 0.03	0.67 ± 0.07	0.50 ± 0.04	0.57 ± 0.01	0.48 ± 0.01
Finkel et al. 2019 n=92, 26/66	all KOs	0.38 ± 0.12	0.63 ± 0.16	0.27 ± 0.15	0.37 ± 0.18	0.47 ± 0.07	0.69 ± 0.04
	filtered KOs	0.36 ± 0.10	0.65 ± 0.07	0.18 ± 0.20	0.25 ± 0.24	0.38 ± 0.09	0.68 ± 0.05
	all OGs	0.43 ± 0.23	0.23 ± 0.40	0.33 ± 0.58	0.28 ± 0.48	0.50 ± 0.00	0.70 ± 0.00
	filtered OGs	0.28 ± 0.00	–	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00	0.72 ± 0.00
	ESM-2	0.35 ± 0.02	0.61 ± 0.05	0.27 ± 0.02	0.37 ± 0.01	0.41 ± 0.01	0.70 ± 0.01
	Bacformer	0.40 ± 0.02	0.72 ± 0.06	0.28 ± 0.00	0.40 ± 0.00	0.42 ± 0.03	0.70 ± 0.03
	DNABERT-S	0.47 ± 0.04	0.72 ± 0.03	0.43 ± 0.12	0.53 ± 0.09	0.49 ± 0.02	0.72 ± 0.02
Lebeis et al. n=35, 15/20	all KOs	0.43 ± 0.00	–	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00	0.57 ± 0.00
	filtered KOs	0.43 ± 0.00	–	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00	0.57 ± 0.00
	all OGs	0.47 ± 0.11	0.20 ± 0.34	0.33 ± 0.58	0.25 ± 0.43	0.50 ± 0.00	0.59 ± 0.00
	filtered OGs	0.41 ± 0.00	–	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00	0.59 ± 0.00
	ESM-2	0.50 ± 0.04	0.59 ± 0.06	0.42 ± 0.06	0.48 ± 0.05	0.47 ± 0.05	0.59 ± 0.05
	Bacformer	0.48 ± 0.02	0.56 ± 0.02	0.35 ± 0.03	0.43 ± 0.03	0.53 ± 0.04	0.58 ± 0.04
	DNABERT-S	0.37 ± 0.05	0.44 ± 0.05	0.38 ± 0.03	0.41 ± 0.03	0.37 ± 0.01	0.56 ± 0.02
OOD test set n=110, 88/22	all OGs	0.60 ± 0.35	0.07 ± 0.12	0.33 ± 0.58	0.11 ± 0.19	0.50 ± 0.00	0.20 ± 0.00
	filtered OGs	0.80 ± 0.00	–	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00	0.20 ± 0.00

Table 4 Additional external data and OOD test set performance. Performance of the same multilayer perceptron architecture on root competence prediction task using different input data, KOs being functional annotations, OGs being clustering-based protein families and ESM-2, Bacformer, and DNABERT-S-derived representations being foundation model based. The performance report is complementing the main table 3, containing either poor or close to random performance, including all the OG-based models and the 3 out of 6 external datasets, Lebeis et al. [43], Finkel et al. 2019 [44], and Duran et al. [45], where all the models failed to generalise. Number of samples, n, in each data set as well as number of negative samples and number of positive samples, neg./pos., is reported. Filters were applied to KOs and OGs to reduce the number of features, such that features with at least 5% variance across genomes are kept.

Input data	Eval.	Threshold	Accuracy	Precision	Recall	F1	AUC	PR-AUC
all KOs	ID	0	0.78	0.74	0.79	0.77	0.78	0.78
all KOs	ID	0.05	0.78	0.74	0.79	0.77	0.84	0.84
all KOs	OOD	0	0.80	0.00	0.00	0.00	0.79	0.43
filtered KOs	ID	0	0.78	0.74	0.79	0.77	0.84	0.84
all OGs	ID	0	0.64	0.59	0.60	0.60	0.69	0.69
filtered OGs	ID	0	0.64	0.59	0.60	0.60	0.68	0.68
ESM-2	ID	0	0.76	0.73	0.78	0.75	0.82	0.82
Bacformer	ID	0	0.77	0.74	0.79	0.76	0.83	0.82
Bacformer	ID	0.05	0.77	0.74	0.78	0.76	0.83	0.78
Bacformer	OOD	0	0.78	0.48	1.00	0.65	0.89	0.60
DNABERT-S	ID	0	0.77	0.72	0.78	0.75	0.84	0.84

Table 5 Performance of a linear baseline on a single test set. Performance of L2 regularised logistic regression on root competence task using different input data, KOs (KEGG Orthology) and OGs (Orthofinder protein families) being annotation- and clustering-based, and ESM-2-, Bacformer-, and DNABERT-S-derived representations being foundation model based. Threshold for binarising the phenotype is reported as well as the type of test set used for the evaluation. Filters were applied to KOs and OGs to reduce the number of features, such that features with at least 5% variance across genomes are kept.

Software	Enviro 1	Enviro 2
Python	3.10.8	3.9.25
PyTorch	2.6.0	2.6.0
Transformers	4.41.0	4.45.1
Biopython	1.83	1.84
h5py	3.14.0	3.14.0
NumPy	1.26.4	1.23.5
pandas	2.3.3	2.3.3
scikit-learn	1.4.1.post1	1.6.1
Optuna	4.0.0	4.4.0
Captum	0.8.0	0.8.0
SciPy	1.15.3	1.13.1
statsmodels	0.14.5	0.13.5
UMAP-learn	0.5.6	-
Matplotlib	3.8.3	3.9.4
seaborn	0.13.2	0.12.1
tqdm	4.67.1	4.64.1

Table 6 Software versions in the computational environments. Versions were retrieved from the two Python environments used for the analysis. Bacformer was manipulated with after the end of the project; the historical package version could not be recovered from the current environment.

Supplementary Figures

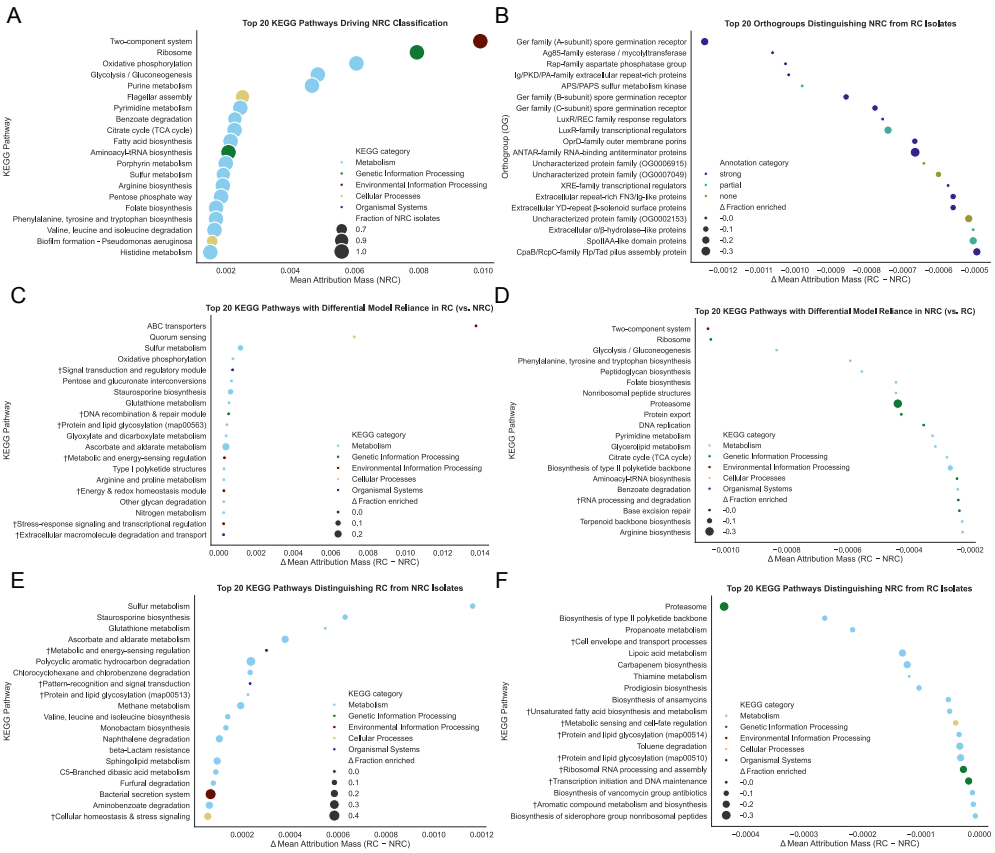


Fig. 7 Additional model-derived functional signals of root competent (RC) and non-root competent (NRC) genomes across KEGG pathways and Orthogroups. Explainability analysis for the best generalising model trained on Bacformer-derived representations. Protein contributions to model predictions were quantified using Integrated Gradients (IGs). Proteins within the top 50% of attribution scores were grouped into KEGG pathways or OGs (Orthofinder protein families). For each group, we calculated two feature-importance metrics: global mean attribution mass (Equation 5) and prevalence, defined as the fraction of genomes containing at least one retained top-attribution protein from that group. (A) Top 20 KEGG pathways associated with NRC classification, ordered by mean attribution mass, consistent with the class specific analysis criteria (Section 5.5.1). (B) Top 20 OGs distinguishing NRC from RC genomes, ordered by Δ mean attribution mass (RC-NRC), consistent with the level 2 relative analysis criteria (Equation 9). Gs are coloured by annotation category: strong, containing at least three KO-annotated proteins; partial, containing at least five otherwise annotated proteins or fewer than three KO annotations; and none, containing fewer than five otherwise annotated proteins. Group names represent consensus functional labels derived from available annotations. (C-D) Top 20 KEGG pathways with differential model reliance in RC (C) or NRC (D) vs. the other class, ordered by Δ mean attribution mass (RC-NRC), consistent with the level 1 relative analysis criteria (Equation 8). (E-F) Top 20 KEGG pathways distinguishing RC (E) or NRC (F) from the other class, ordered by Δ mean attribution mass (RC-NRC), consistent with the level 2 relative analysis criteria (Equation 9). KEGG pathways marked with † mapped to non-bacterial pathways and were renamed to carry organism agnostic, molecularly mechanistic name based on its constituent KOs.

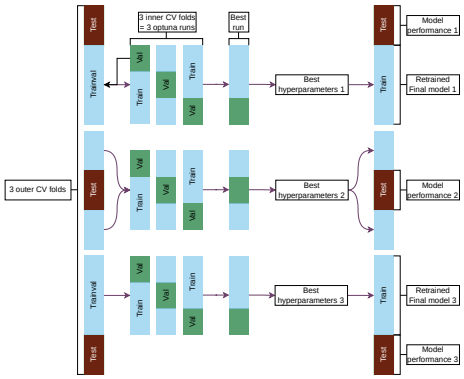


Fig. 8 Illustration of the cross-validation set up used in this study. 3 inner 3 outer fold cross-validation. The inner folds are used for hyperparameter tuning and the outer folds for testing. Within each outer fold, the dataset was split into a test set and a training validation (trainval) set that is further split into training (train) and validation (val) set in the inner folds. After the inner folds, the model was retrained using the best hyperparameters on the training (train) data previously called trainval. We obtain 3 separate models, and 3 test set performance evaluations.

Supplementary Hyperparameters

The model hyperparameters used for training are reported in the repository with one file per model containing all necessary configuration for reproducibility.